
Quantitative Research Report

Robust Portfolio Allocation and Distributional Volatility Forecasting

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Two reproducible empirical studies connect statistical uncertainty estimates to portfolio decisions. The first estimates directional covariance uncertainty through circular-block bootstrap resampling and incorporates the resulting upper covariance matrix into a capped long-only minimum-variance problem. The second combines forward-variance experts through bounded-loss Hedge aggregation, calibrates an upper variance forecast on non-overlapping residual streams, and converts the forecast into a transaction-cost-aware SPY exposure rule.

The report emphasizes temporal integrity, explicit benchmark construction, dependence-aware inference, and reproducibility from immutable public-data snapshots.

Abstract

This report develops two quantitative methods for risk estimation and portfolio control. The allocation study aggregates the Kenneth French 100 size–book-to-market portfolios into a continuous 25-portfolio panel using lagged capitalization weights. At each monthly rebalance, circular-block bootstrap resampling estimates dispersion in the sample eigenvector directions. Directional upper quantiles generate a positive-semidefinite covariance matrix for an exact capped long-only quadratic program. Hyperparameters are selected on 2000–2012 and evaluated over 2013–2026. The resulting portfolio records 16.45% annualized volatility, compared with 20.15% for equal weight and 16.46% for validation-selected Ledoit–Wolf GMV. The relative volatility reduction versus equal weight is 18.39%, with a paired block-bootstrap 95% interval of 14.95–22.58%. Performance is essentially tied with Ledoit–Wolf, while the bootstrap estimator provides an explicit directional decomposition of covariance uncertainty.

The volatility study forecasts annualized variance over the next 22 trading days. Labels enter estimation only after their complete target windows have matured. Six experts are combined quarterly through bounded-loss Hedge aggregation, and a 90% upper forecast is calibrated across 22 interlaced residual streams with non-overlapping target windows within each stream. Over 2013–2026, the ensemble reduces QLIKE by 3.34% and log-RMSE by 14.94% relative to raw VIX variance. The corresponding HAC p -values are 0.608 for QLIKE loss differences and 3.20×10^{-15} for squared log-error differences. Empirical upper-bound coverage is 89.03%. The upper-forecast controller reaches 8.30% annualized volatility and an 11.56% maximum drawdown, compared with 16.81% and 33.72% for SPY buy-and-hold; standard VIX targeting retains the stronger risk-adjusted return.

Key findings

Study	Evaluation evidence
Robust allocation	16.45% annualized volatility versus 20.15% for equal weight, with performance aligned with Ledoit–Wolf GMV.
Distributional volatility	14.94% lower log-RMSE than raw VIX variance and 89.03% upper-bound coverage; the controller reduces maximum drawdown to 11.56%.

1 Research design

1.1 Objectives

The two studies follow the same quantitative architecture: estimate uncertainty, preserve temporal causality, and embed the estimate directly in the decision rule. In the allocation model, bootstrap dispersion modifies the covariance matrix before optimization. In the volatility model, matured forecast errors determine both expert weights and the upper variance forecast used for exposure control.

The empirical design combines fixed development and validation periods, explicit transaction-cost accounting, dependence-aware inference, benchmark portfolios, and machine-checkable temporal audits.

1.2 Experimental chronology

Table 1: Chronology used by the two studies.

Stage	Allocation study	Volatility study	Role in the empirical workflow
Historical estimation	Through 1999	Through 2004	Initial rolling estimates and feature construction
Development	–	2005–2008	Ridge-expert specification and QLIKE loss cap
Validation	2000–2012	2009–2012	Hyperparameter selection and calibration design
Evaluation	2013–2026	2013–2026	Fixed-model comparison with benchmark strategies

All model updates, portfolio optimizations, and residual calibrations use information available at the decision date. The final specification is reported separately over 2013–2026 after development and validation choices have been fixed.

1.3 Data and accounting conventions

The allocation study uses the Kenneth French daily 100 size–book-to-market portfolio archive. The volatility study uses SPX closes, Cboe VIX history, SPY adjusted-close total returns, and the FRED three-month Treasury constant-maturity yield. Raw snapshots and SHA-256 hashes are included in the accompanying repository.

Portfolio returns include explicit turnover costs. In the volatility strategy, uninvested capital earns the Treasury proxy and leverage above one is financed at the same rate. This common-rate financing convention isolates the forecasting and exposure mechanisms from broker-specific funding schedules.

2 Bootstrap-calibrated distributionally robust allocation

2.1 Construction of a continuous characteristic panel

The source archive contains 100 value-weighted portfolios on a 10×10 size–book-to-market grid. Extreme cells may be empty for an entire portfolio-formation year and are represented by missing-value codes. Dropping every day on which any cell is empty would introduce artificial gaps. Instead, adjacent 2×2 cells are aggregated into 25 portfolios.

Let $R_{i,t}$ denote the return of cell i , $N_{i,t}$ its firm count, and $S_{i,t}$ its reported average firm size. For an aggregated group g , the implemented return is

$$R_{g,t} = \frac{\sum_{i \in g: R_{i,t} \text{ observed}} M_{i,t-1} R_{i,t}}{\sum_{i \in g: R_{i,t} \text{ observed}} M_{i,t-1}}, \quad M_{i,t-1} = N_{i,t-1} S_{i,t-1}. \quad (1)$$

Empty cells receive zero weight. The capitalization measure is lagged, so same-day returns do not affect the aggregation weights. The result is a complete 25-column panel from July 1963 through May 2026.

2.2 Directional bootstrap ambiguity set

At a rebalance date t , let $X_t \in \mathbb{R}^{L \times p}$ be the trailing return matrix and write the sample covariance as

$$S_t = U_t \text{diag}(\lambda_{1,t}, \dots, \lambda_{p,t}) U_t^\top. \quad (2)$$

Circular moving-block resamples preserve short-range temporal dependence. For each bootstrap draw b , the resampled data are projected onto the *fixed* sample eigenvectors, producing directional variances $\lambda_{j,t}^{*(b)}$. For quantile level q ,

$$\bar{\lambda}_{j,t}(q) = \max \left\{ \lambda_{j,t}, Q_q(\lambda_{j,t}^{*(1)}, \dots, \lambda_{j,t}^{*(B)}) \right\}, \quad (3)$$

and

$$\bar{\Sigma}_t(q) = U_t \text{diag}(\bar{\lambda}_{1,t}, \dots, \bar{\lambda}_{p,t}) U_t^\top. \quad (4)$$

This matrix is the worst element, for every feasible long-only portfolio, in the rectangular set

$$\mathcal{U}_t(q) = \{ U_t \text{diag}(d) U_t^\top : \lambda_{j,t} \leq d_j \leq \bar{\lambda}_{j,t}(q) \}. \quad (5)$$

The set is positive semidefinite by construction and inflates estimated variance in directions supported by bootstrap dispersion. The specification isolates eigenvalue uncertainty in the sample eigenspace, producing a directional and economically interpretable risk adjustment.

2.3 Optimization and trading rule

Let w_t^- be the pre-trade portfolio obtained by drifting the previous target weights through realized returns. The monthly target solves

$$\min_{w \in \mathbb{R}^p} w^\top \frac{\hat{\Sigma}_t}{\text{tr}(\hat{\Sigma}_t)/p} w + \gamma \|w - w_t^-\|_2^2, \quad (6)$$

$$\text{subject to } \mathbf{1}^\top w = 1, \quad 0 \leq w_i \leq c. \quad (7)$$

The trace normalization fixes the scale of the displacement penalty across covariance estimates. A specialized active-set solver is used; the supplied test suite compares its objective value with a generic SLSQP solution to tolerance 10^{-9} .

The validation grid is $L \in \{252, 504\}$, $q \in \{0.75, 0.90, 0.95\}$, $c \in \{0.10, 0.20\}$, and $\gamma \in \{0, 0.5, 2\}$. The unique objective is minimum net annualized volatility over 2000–2012. The selected configuration is

$$L = 252, \quad q = 0.75, \quad c = 0.20, \quad \gamma = 0,$$

with 21-day blocks and 128 bootstrap replications at each rebalance. The validation criterion selects $\gamma = 0$, so turnover is controlled through monthly rebalancing, weight caps, and net-of-cost model selection rather than an explicit displacement penalty.

At each rebalance, the strategy pays

$$C_t = 10^{-3} \sum_{i=1}^p |w_{i,t}^{\text{target}} - w_{i,t}^-|, \quad (8)$$

which is 10 basis points per dollar traded under the full L^1 weight change convention.

2.4 Evaluation results

Table 2: Allocation results, 2 January 2013–29 May 2026. Costs are 10 bps per dollar traded.

Strategy	CAGR	Volatility	Sharpe	Max drawdown	Annual L^1 turnover
Bootstrap upper covariance	12.90%	16.45%	0.820	−36.51%	1.26
Ledoit–Wolf GMV	12.87%	16.46%	0.818	−36.68%	1.25
Sample GMV	12.79%	16.46%	0.814	−36.66%	1.23
Equal weight	13.12%	20.15%	0.713	−41.26%	0.22
Inverse volatility	13.28%	19.67%	0.733	−40.75%	0.29

The primary portfolio lowers annualized volatility by 18.39% relative to equal weight. A paired circular moving-block bootstrap, resampling candidate and benchmark returns jointly with 21-day blocks and 5,000 replications, gives a 95% interval of 14.95–22.58%. The Sharpe difference relative to equal weight is positive but less decisive: the corresponding interval is -0.052 to 0.290 .

Relative to validation-selected Ledoit–Wolf GMV, the volatility reduction is 0.11%, with a paired 95% interval of 0.01–0.23%. The two methods therefore occupy the same empirical risk region. The bootstrap construction adds a transparent decomposition of the covariance adjustment across sample eigenvector directions.

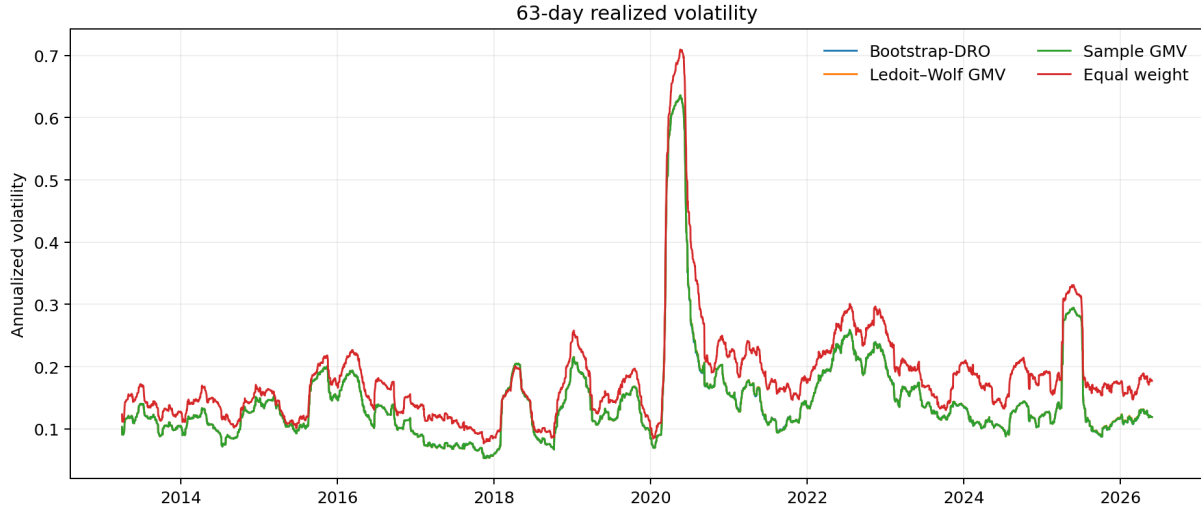


Figure 1: Rolling 63-day annualized volatility. The three covariance-based portfolios remain close to one another and below equal weight for most of the evaluation period.

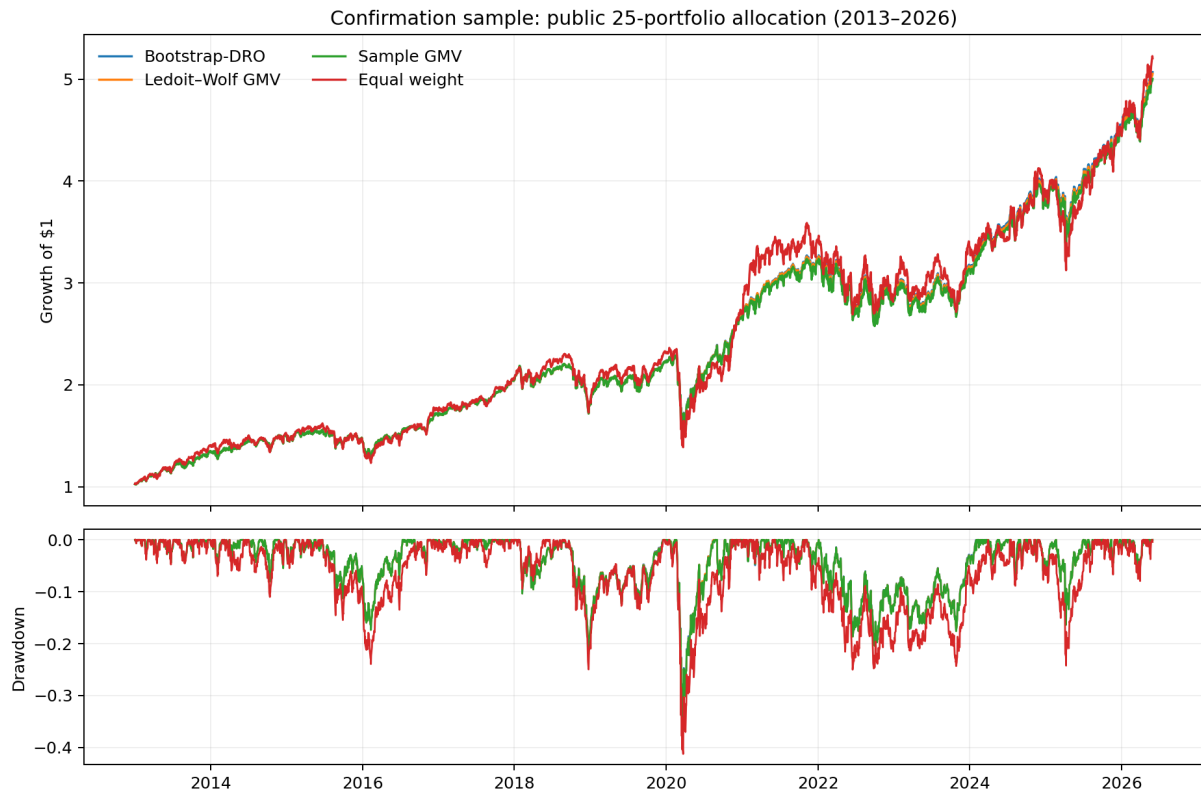


Figure 2: Cumulative wealth and drawdown for the allocation strategies. The covariance-based portfolios primarily improve the risk path, while equal weight reaches comparable terminal wealth through higher market exposure.

2.5 Sensitivity and risk calibration

Changing the estimator block length from 5 to 63 days and the number of bootstrap replications from 64 to 256 leaves evaluation volatility between 16.44% and 16.45%. The mean L^1 distance from the primary target weights remains below 0.013. At 50 bps per dollar traded, CAGR falls from 12.90% to 12.33%, while volatility is essentially unchanged.

The mean next-month realized-to-predicted volatility ratio is 0.941 for the bootstrap covariance and 0.998 for Ledoit–Wolf. The directional upper covariance is conservative on average. Forecast dispersion remains substantial during abrupt volatility transitions, highlighting the value of dynamic regime information beyond a static monthly covariance estimate.

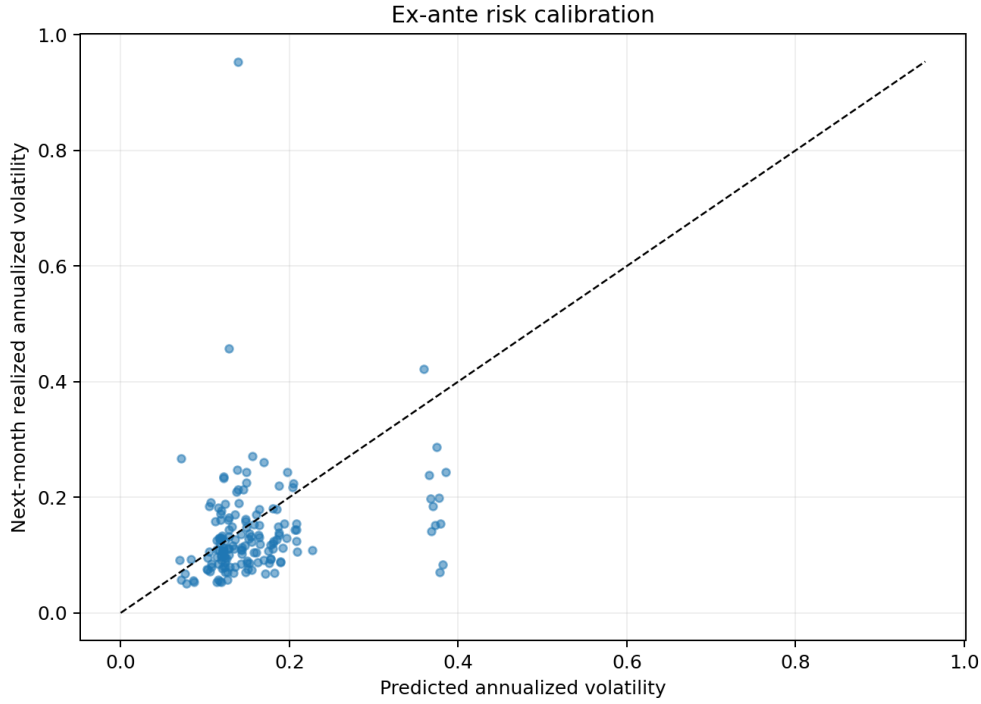


Figure 3: Ex-ante portfolio-risk forecasts and next-month realized volatility, with the 45-degree line shown as the calibration reference.

3 Maturity-safe distributional volatility forecasting

3.1 Target and information set

The target at forecast origin t is annualized variance over the next 22 trading days,

$$RV_{t,+}^{(22)} = \frac{252}{22} \sum_{j=1}^{22} r_{t+j}^2. \quad (9)$$

Every origin stores the final date of its target window. A historical label from origin s may enter a model refit or residual calibration at date t only when

$$\text{target_end}(s) < t. \quad (10)$$

The strict inequality prevents the current close from entering a target used at that same close. This restriction is enforced in emitted audit files and in unit tests.

3.2 Expert forecasts and bounded-loss Hedge

The forecast panel contains six experts:

1. ridge regression using log VIX variance;

2. ridge regression using VIX and HAR-style 5-, 22-, and 66-day variance features;
3. ridge regression adding downside-return and VIX-change features;
4. raw VIX variance;
5. trailing 22-day variance;
6. EWMA variance with decay 0.94.

Ridge penalties and Duan smearing factors are selected for each regression expert on 2005–2008. Models are then refit quarterly on expanding samples of fully matured labels.

For realized variance y and forecast $\hat{y}$, the QLIKE loss is

$$\ell(y, \hat{y}) = \frac{y}{\hat{y}} - \log\left(\frac{y}{\hat{y}}\right) - 1. \quad (11)$$

The loss cap $L = 4.738868$ is fixed as the 99th percentile of pooled expert losses on the development period. At each quarterly update, the latest $n \leq 504$ matured observations are used. With $K = 6$ experts,

$$\tilde{\ell}_{k,s} = \min\{\ell_{k,s}, L\}, \quad \eta_t = \sqrt{\frac{8 \log K}{nL^2}}, \quad (12)$$

$$\pi_{k,t} = \frac{\exp\left(-\eta_t \sum_s \tilde{\ell}_{k,s}\right)}{\sum_{j=1}^K \exp\left(-\eta_t \sum_s \tilde{\ell}_{j,s}\right)}, \quad \hat{v}_t = \sum_{k=1}^K \pi_{k,t} \hat{v}_{k,t}. \quad (13)$$

The two-year memory allows the ensemble to adapt across volatility regimes. The learning rate is determined analytically by the bounded-loss Hedge expression.

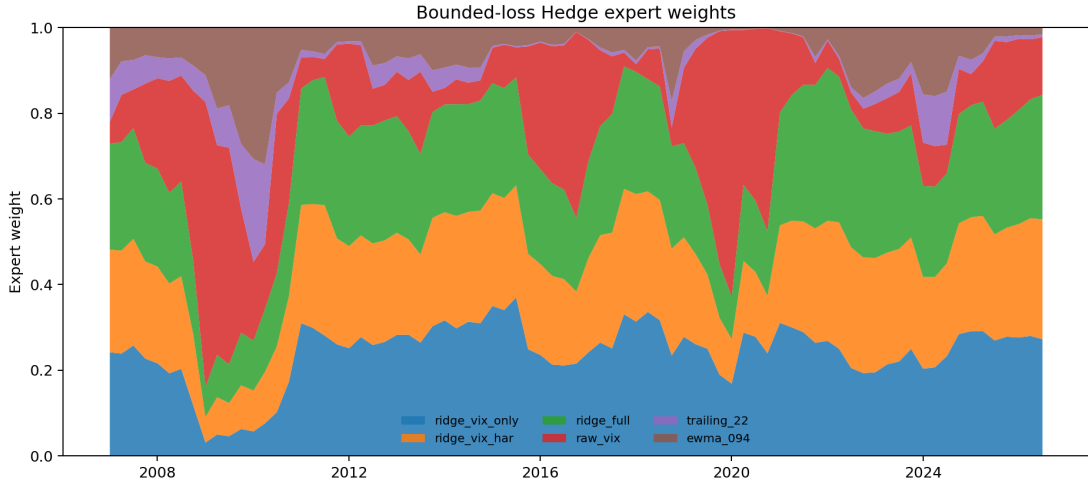


Figure 4: Quarterly Hedge weights. The aggregation rule reallocates weight gradually as relative expert performance changes.

3.3 Interlaced upper-forecast calibration

Define the log residual

$$e_s = \log RV_{s,+}^{(22)} - \log \hat{v}_s. \quad (14)$$

Forecast origins are divided into 22 streams according to their index modulo 22. Consecutive origins in a given stream are 22 trading days apart, so their forward target windows do not overlap. At origin t , calibration uses at most the latest 63 matured residuals in the same stream:

$$h(s) = h(t), \quad \text{target_end}(s) < t. \quad (15)$$

The 90% upper forecast is

$$U_t = \hat{v}_t \exp\{Q_{0.90}(e_s)\}. \quad (16)$$

The interlaced design is selected as the primary calibration method because every within-stream residual sample is built from disjoint forward windows. A pooled residual estimator achieves lower validation pinball loss, while the interlaced estimator offers cleaner dependence control for horizon-overlapping targets.

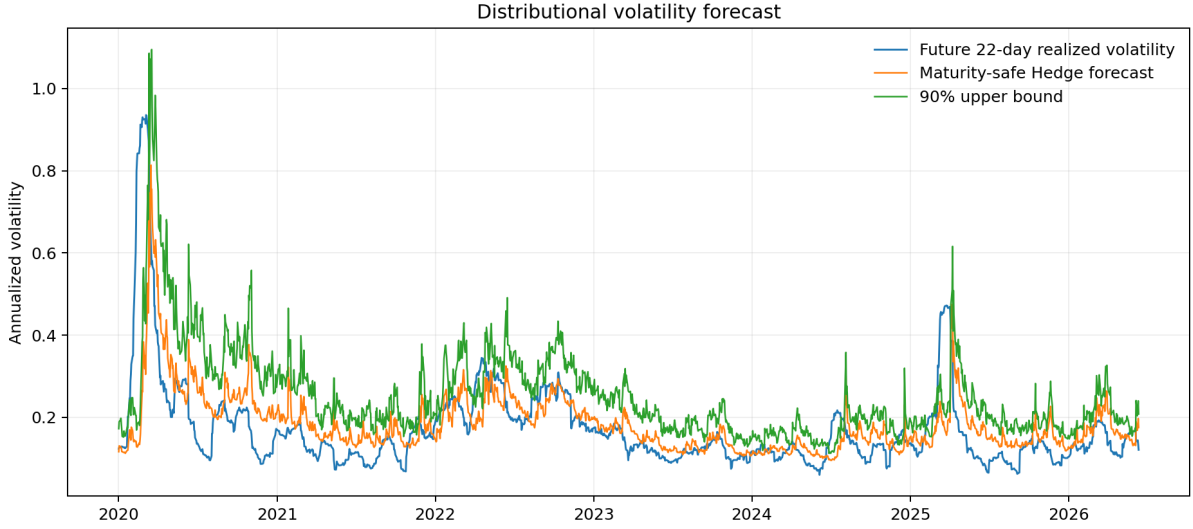


Figure 5: Point forecasts, realized forward variance, and the calibrated 90% upper forecast. The upper series widens sharply in stressed periods.

3.4 Forecast evaluation

Table 3: Forecast performance on 3,381 evaluation origins. Lower QLIKE and log-RMSE are better.

Forecast	QLIKE	Log-RMSE	Log-MAE	Level correlation
Hedge ensemble	0.3820	0.7755	0.6140	0.4614
Raw VIX variance	0.3952	0.9117	0.7647	0.4542
Trailing 22-day variance	0.6419	0.9078	0.6994	0.2997

Relative to raw VIX variance, QLIKE falls by 3.34%. A Newey–West test with 21 lags yields $p = 0.608$ for the mean QLIKE-loss difference. Log-RMSE falls by 14.94%, and the HAC test on squared log-error differences yields $p = 3.20 \times 10^{-15}$. The ensemble’s strongest improvement is therefore in multiplicative forecast accuracy rather than average QLIKE loss.

Overall upper-bound coverage is 89.03% at a nominal 90%, across 3,381 evaluable forecasts. Coverage equals 86.58% in the lowest validation-defined VIX quintile and ranges from 90.66% to 95.68% in the middle three quintiles. The aggregate calibration is close to nominal, with systematic variation across volatility regimes.

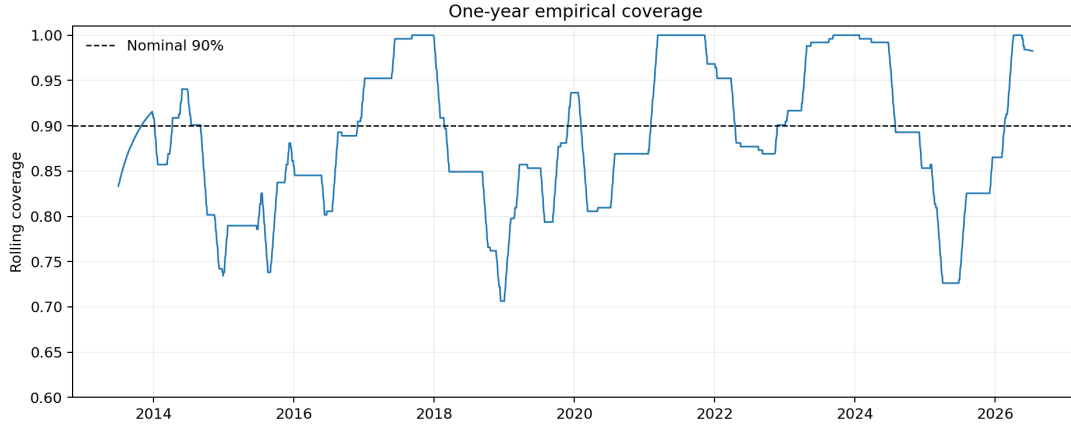


Figure 6: Rolling one-year empirical coverage. Calibration varies across market regimes while remaining close to the 90% target in aggregate.

3.5 Economic decision rule

At the close of day t , the next-day SPY exposure is

$$\ell_t = \text{clip} \left(\frac{0.12}{\sqrt{U_t}}, 0.2, 1.5 \right). \quad (17)$$

Let R_{t+1}^{SPY} be the next adjusted-close total return and R_t^f the daily Treasury proxy. With one basis point per unit change in exposure,

$$R_{t+1}^p = \ell_t R_{t+1}^{\text{SPY}} + (1 - \ell_t) R_t^f - 10^{-4} |\ell_t - \ell_{t-1}|. \quad (18)$$

Benchmarks include buy-and-hold, standard VIX targeting, point-forecast targeting, constant exposure fixed from validation, and a VIX rule scaled using validation-period mean exposure. An equal-exposure VIX series isolates timing effects from average market exposure.

Table 4: SPY risk-control results, 2 January 2013–14 July 2026. Sharpe is measured over the cash proxy.

Strategy	CAGR	Volatility	Sharpe over cash	Max drawdown	Average exposure	Annual turnover
Distributional upper bound	8.76%	8.30%	0.839	−11.56%	0.665	10.33
Standard VIX target	9.90%	9.33%	0.867	−13.89%	0.745	9.67
Point-forecast target	10.91%	10.84%	0.846	−15.56%	0.871	9.94
Fixed validation exposure	7.52%	7.12%	0.804	−15.33%	0.424	0.00
VIX, validation-matched exposure	8.09%	7.23%	0.866	−10.58%	0.573	7.42
SPY buy-and-hold	14.88%	16.81%	0.804	−33.72%	1.000	0.00

The distributional controller reduces maximum drawdown by 22.16 percentage points relative to buy-and-hold and by 3.77 points relative to the fixed validation exposure. Its average exposure is 0.665, so the comparison with buy-and-hold reflects both risk scaling and timing. Standard VIX targeting records a higher Sharpe over cash, while the validation-matched VIX rule also achieves a slightly smaller drawdown. At equal average exposure, the distributional rule improves maximum drawdown by 0.79 percentage points. The principal economic effect is stronger downside compression around high upper-variance forecasts.

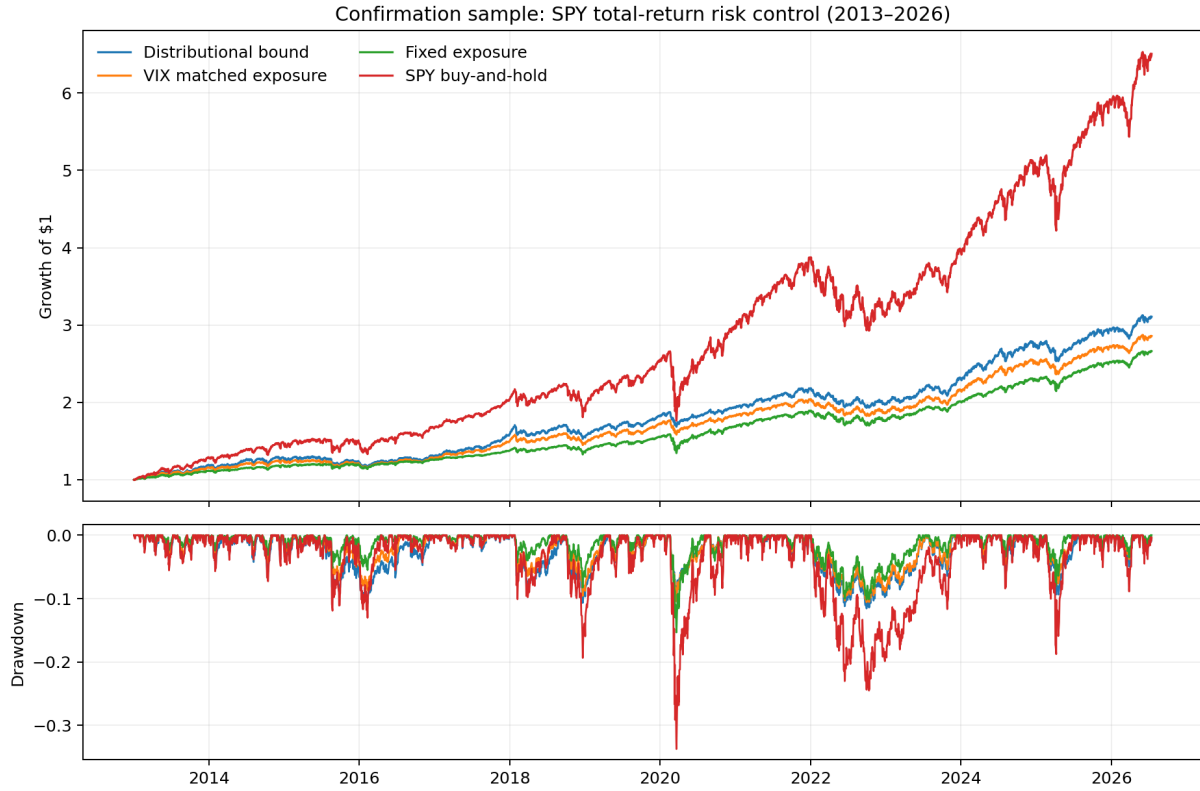


Figure 7: Cumulative wealth and drawdown for the SPY control rules. Buy-and-hold compounds faster because it carries full exposure; the controlled strategies trade return for lower risk.

4 Reproducibility and audit controls

The repository regenerates all reported tables and figures from immutable public-data snapshots in a Python 3.12 environment through a single execution script. The audit suite verifies:

- SHA-256 verification of all five raw data snapshots;
- feasibility tests for capped simplex projections and selected portfolio weights;
- an active-set QP objective comparison against SLSQP at 10^{-9} tolerance;
- positive-semidefinite and trace-inflation checks for bootstrap covariance matrices;
- continuity and completeness checks for the 25-portfolio panel;
- strict future-target and label-maturity checks;
- overlap checks requiring calibration observations to come from the same interlaced stream with disjoint target windows;
- accounting identities for all-cash and all-SPY strategies;
- non-empty artifact checks for every generated output.

All ten integrity tests pass, and the raw-data manifest verifies every snapshot without a hash mismatch. The tests cover data continuity, optimization feasibility, temporal causality, calibration overlap, accounting identities, and artifact generation.

5 Scope and research extensions

5.1 Allocation study

The current allocation experiment uses academic characteristic portfolios, which provide a long and internally consistent panel for comparing covariance estimators. A constituent-level implementation would add security-specific spreads, impact, capacity, taxes, and point-in-time membership. On the statistical side, the present

ambiguity set perturbs eigenvalues in a fixed sample eigenspace. Joint eigenvalue-eigenvector uncertainty and structural-break adaptation define the next extension.

A natural refinement is a simultaneous confidence region for portfolio variance over the capped simplex, replacing coordinatewise directional quantiles with a set calibrated directly to uniform risk underestimation. Constituent-level point-in-time data would then allow the displacement penalty and rebalance frequency to be selected under nonlinear execution costs.

5.2 Volatility study

The volatility experiment uses daily squared returns as the forward-variance proxy. Intraday realized kernels, HARQ measurement-error corrections, option-surface features, and point-in-time variance-risk-premium measures would sharpen both the target and the feature set. The interlaced construction controls overlap within calibration streams; regime-conditioned residual models offer the next step toward conditional coverage.

An execution-oriented extension would implement exposure through liquid index futures with observed funding, bid–ask spreads, and roll costs. A pre-specified evaluation on an additional market or forward period would then measure portability of the expert weights, calibration quantiles, and exposure mapping.

6 Conclusion

The allocation study integrates bootstrap-estimated directional uncertainty into an exact convex portfolio problem. The resulting portfolio reduces annualized volatility by 18.39% relative to equal weight and matches the risk performance of Ledoit–Wolf GMV while preserving an interpretable decomposition of covariance uncertainty.

The volatility study combines maturity-safe expert aggregation with non-overlapping residual calibration. The ensemble materially improves log-scale forecast accuracy relative to raw VIX variance, and the calibrated upper forecast supports an exposure rule with substantially lower volatility and drawdown than buy-and-hold. VIX targeting remains the stronger benchmark on risk-adjusted return, locating the contribution of the proposed method in distributional forecasting and downside-sensitive control.

Together, the studies demonstrate a unified quantitative workflow in which uncertainty is estimated explicitly, audited for temporal validity, and carried through to portfolio decisions with reproducible benchmarks and transaction-cost accounting.

A Selected configurations and sensitivity ranges

Table 5: Primary allocation configuration.

Component	Selected value
Lookback	252 trading days
Bootstrap block length	21 trading days
Bootstrap replications	128 per rebalance
Directional quantile	0.75
Maximum portfolio weight	0.20
Displacement penalty	0
Rebalance frequency	Monthly
Transaction cost	10 bps per dollar traded
Validation objective	Minimum net annualized volatility, 2000–2012

Table 6: Primary volatility configuration.

Component	Fixed value
Forecast horizon	22 trading days
Expert count	6
Hedge memory	Up to 504 matured forecasts
QLIKE loss cap	4.738868, fixed on 2005–2008
Model refit frequency	Quarterly
Calibration streams	22
Scores per stream	Up to 63
Nominal upper quantile	0.90
Target volatility	12% annualized
Exposure range	0.2 to 1.5
Transaction cost	1 bp per unit exposure change

B Additional diagnostics

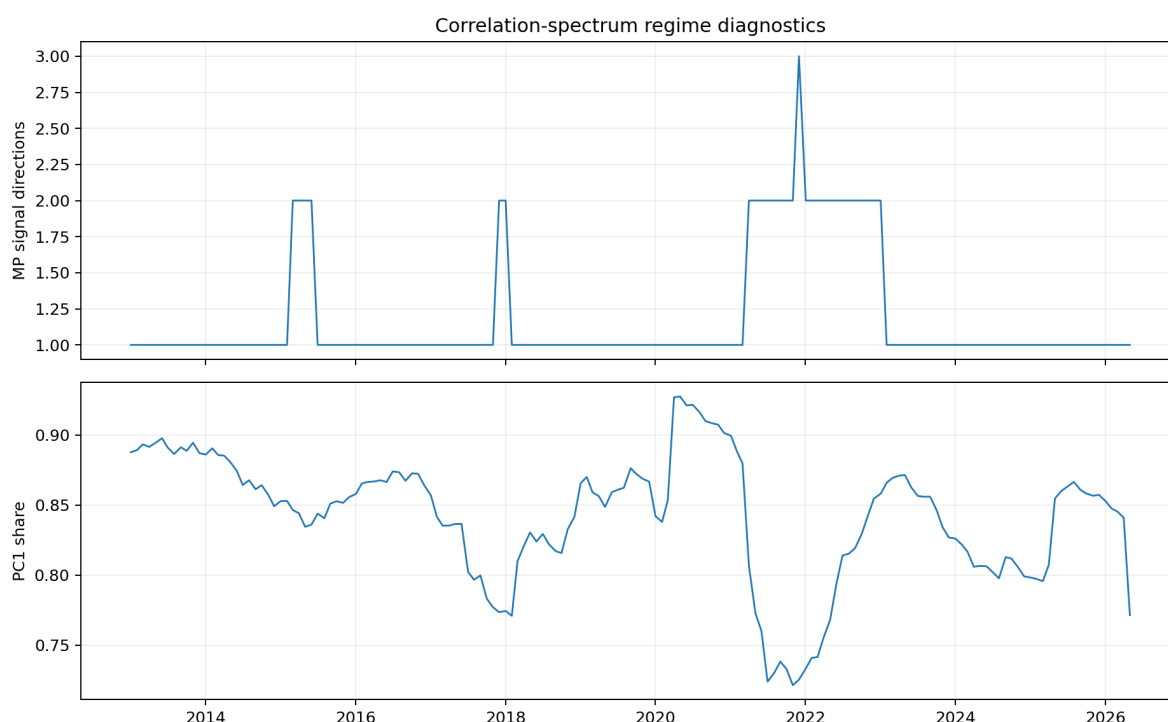


Figure 8: Correlation-spectrum diagnostics for the allocation study. The upper panel records the number of sample eigenvalues above the Marchenko–Pastur reference edge; the lower panel shows the trace share associated with those directions.

C Public data sources

- Kenneth French Data Library, daily 100 portfolios formed on size and book-to-market: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
- Cboe VIX historical data: https://www.cboe.com/tradable_products/vix/vix_historical_data
- FRED DGS3MO, three-month Treasury constant-maturity rate: <https://fred.stlouisfed.org/series/DGS3MO>
- Yahoo Finance chart snapshots for SPX and SPY, stored in the repository as immutable JSON files.

Research report prepared from the public-data repository and its generated audit artifacts.
All results are generated from the accompanying reproducible research repository.